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The two dimensional microstructure characterization of cemented carbides with an automatic image analysis process

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ABSTRACT

The traditional two dimensional microstructure characterization of cemented carbide, based on stereology of linear intercept method, requires tedious and subjective manual measurements. In this study, an automatic image analysis procedure with two key techniques, i.e. maximum class square error method and watershed transformation method, has been successfully developed. The image analysis for WC-16Co cemented carbides with this procedure easily acquires consistent microstructure parameters. The analysis for area weighted WC grain size, as well as the subsequent mean free path of Co binder show quite different results compared with the conventional number weighted data. It is found that for both number weighted and area weighted data, the contiguity of WC/WC grains is insensitive to the variation of either mean WC grain size or mean free path of Co binder. The mean WC grain size is linearly related to the mean free path of Co binder. The hardness of cemented carbide, having a linear relationship with the inverse square root of mean WC grain size, conforms to Chermont and Osterstock's model. Although it is too early to conclude whether number weighted or area weighted WC grain size (and subsequent mean free path of Co binder) is better, this study shows that area weighted WC grain size and the corresponding mean free path of Co binder are more suitable for Chermont and Osterstock's hardness modeling compared with number weighted WC grain size. The area weighted WC grain size and subsequent mean free path of Co binder, which have rarely been considered for microstructure characterization of cemented carbide previously, could be the key parameters for a better understanding of the microstructure evolution, as well as a better mechanical behavior modeling for cemented carbide.

1. Introduction

Cemented carbide, as a special composite of carbide skeleton intertwining with metal binder, has the superior combination of high wear resistance and high toughness over other hard materials such as ceramics, tool steels, as well as super hard materials. Due to its unique mechanical properties, cemented carbide has been used widely in metal cutting, petroleum drilling [1,2], and especially in shield tunnel boring and rock mining [3–5], where simultaneously high wear resistance and high toughness is a must for good performance. Cemented carbide, according to its carbide grain size, can be divided into several categories, i.e. nano cemented carbide, submicron cemented carbide, micron cemented carbide, and coarse-grained cemented carbide. In spite of the varieties in cemented carbide grades, it has been well known that the mechanical behavior of cemented carbide is controlled primarily by its microstructure parameters, such as carbide grain size, mean free path of binder, contiguity of carbide grains, etc. [6–9], which are basically determined by the raw powder, milling and pressing process, sintering temperature, and dwelling time [10].

Currently, both the numerical and finite element modeling has been well established to show the intrinsic relationship between mechanical behavior and two dimensional microstructure parameters of cemented carbide [6–9]. It becomes quite convenient to predict the mechanical properties once the microstructure parameters are available. The two dimensional microstructure characterization of cemented carbide, which is originally based on the stereology theory of linear intercept, has been used in quite a lot research [11–13]. The problem of linear intercept method is that it requires tedious and subjective manual measurements. Although many software and image process techniques have been developed for faster image analysis than linear intercept method [14–24], the automatic image process for binary segmentation, as well as the carbide grain boundary partition of cemented carbide

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Table 1
Nominal composition (wt%) and the size of the raw WC powder in this study.

Sample No.	WC	Co	Nominal size of raw WC powder (μm)
WC-1	84	16	1
WC-3	84	16	3
WC-6	84	16	6
WC-9	84	16	9

microstructure, have not been fully developed yet, making it difficult for automatic microstructure characterization to acquire reliable and consistent results, especially for cemented carbide.

In this study, two key image analysis techniques, i.e. maximum class square error method and watershed transformation method, were introduced to two dimensional automatic microstructure characterization of cemented carbide. An automatic procedure based on these two techniques was developed for microstructure analysis of WC-16Co cemented carbide with four different carbide grain sizes. Automatic and reliable two dimensional microstructure characterization was successfully made based on this procedure. The hardness modeling of WC–16Co was investigated as well in this study.

2. Experimental procedures

2.1. Sample preparation

As shown in Table 1, 4 kinds of cemented carbide with different raw WC particle sizes but the same Co content were prepared for this investigation. WC powder with the FSSS particle size (the average particle size measured by Fisher Sub-Sieve Sizer) of 1, 3, 6, and 9 μm was mixed with Co powder (FSSS particle size of 1 μm), respectively. The powder mixture was pressed and sinter-hipped at 1450 °C for 90 min. The sintered samples were sectioned and polished, and the microstructure was taken in SEM. Automatic microstructure characterization was made to all these samples with a specially developed automatic image analysis procedure including two key image analysis techniques, i.e. maximum class square error method and watershed transformation method.

2.2. Microstructure binarization

Binary segmentation of image is the fundamental of microstructure characterization not only for cemented carbide but for all the materials. Currently, the manual binarization of cemented carbide still heavily depends on the operator's experience, which can introduce significant subjective error. In this study, maximum class square error method proposed by Otsu [25] was used to minimize the subjective error during image binary segmentation process. Per Otsu's theory, the optimum threshold grayscale value to binarize a gray scale microstructure with two phases can be determined by [25]:

$$t^{*} = \mathrm{ArgMax}_{0\le i\le G-1}[P_A(\omega_A-\omega_0)^2 + P_B(\omega_B-\omega_0)^2] \quad (1)$$

where G is the maximum grayscale, P_A and P_B are the probabilities of all the pixels with gray level less than or more than a specific threshold grayscale value t, respectively, ω_A and ω_B are the average grayscale of all the pixels with gray level less than or more than the grayscale value t, respectively, and ω_0 is the average grayscale of the overall microstructure.

2.3. WC/WC grain partition

WC/WC grain boundary determination is critical to evaluate WC grain size and contiguity of carbide. In this study, classical watershed algorithm was used to partition the WC/WC grains effectively and accurately. The watershed procedure starts with the distance transformation of the binarized microstructure, which transforms the WC particles (black regions in the binarized microstructure) into gray scale regions where the gray scale of each pixel in WC particles represents the distance between this pixel to the nearest white pixel, i.e. the Co binder region in the binarized microstructure. Then the distance transformation image was binarized into a black and white image where the innermost regions of WC particles become black and surrounded by white matrix. Finally, watershed process was made to this binarized distance transformation image to evenly dilate the black regions pixel by pixel until neighbor black regions meet each other and the pixels of encounter form the partition lines between WC/WC grains. It was found that both the good denoising and binarization of the original microstructure beforehand is critical to the subsequent accurate watershed process without over or under partition of WC/WC grains.

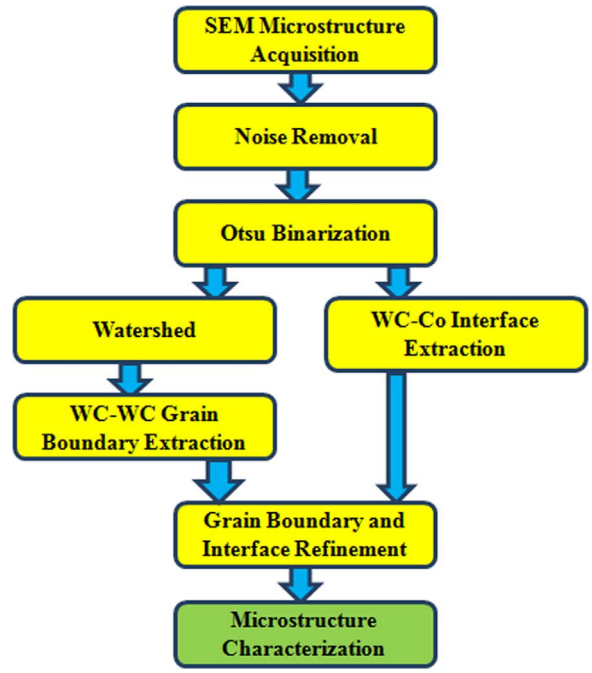

Fig. 1. The automatic microstructure characterization procedure developed in this study.

2.4. Automatic microstructure characterization for cemented carbide

Based on the above mentioned microstructure processing techniques, this study developed an automatic microstructure characterization procedure as shown in Fig. 1. The procedure starts with SEM microstructure acquisition. After smoothing and noise removal, the microstructure binarization was made. Then the watershed was used to partition the WC/WC grains. Finally, the microstructure characterization was made based on the well binarized and partitioned image.

3. Results and discussion

3.1. Image processing

Fig. 2 shows the original microstructure of cemented carbides in Table 1. With the increasing of raw WC particle size, the average WC grain size increases after sintering. For WC-1 and WC-3, certain amount of coarse WC grains coexist with very fine WC grains due to the lack of carbide grain growth inhibitors (such as Cr_3C_2 and VC) in the initial powder mixture.

Fig. 3 shows the grayscale histogram of four images in Fig. 2. According to maximum class square error theory in Eq. (1), the

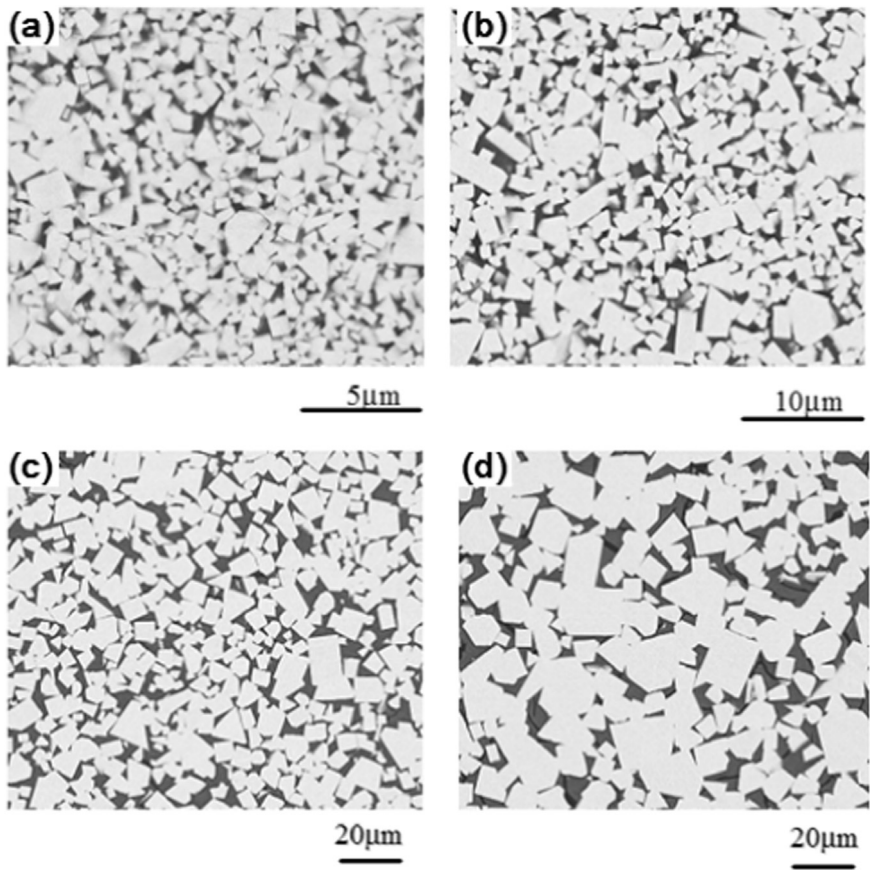

Fig. 2. Backscattered electron SEM microstructure for samples in this study: (a) WC-1, (b) WC-3, (c) WC-6, and (d) WC-9.

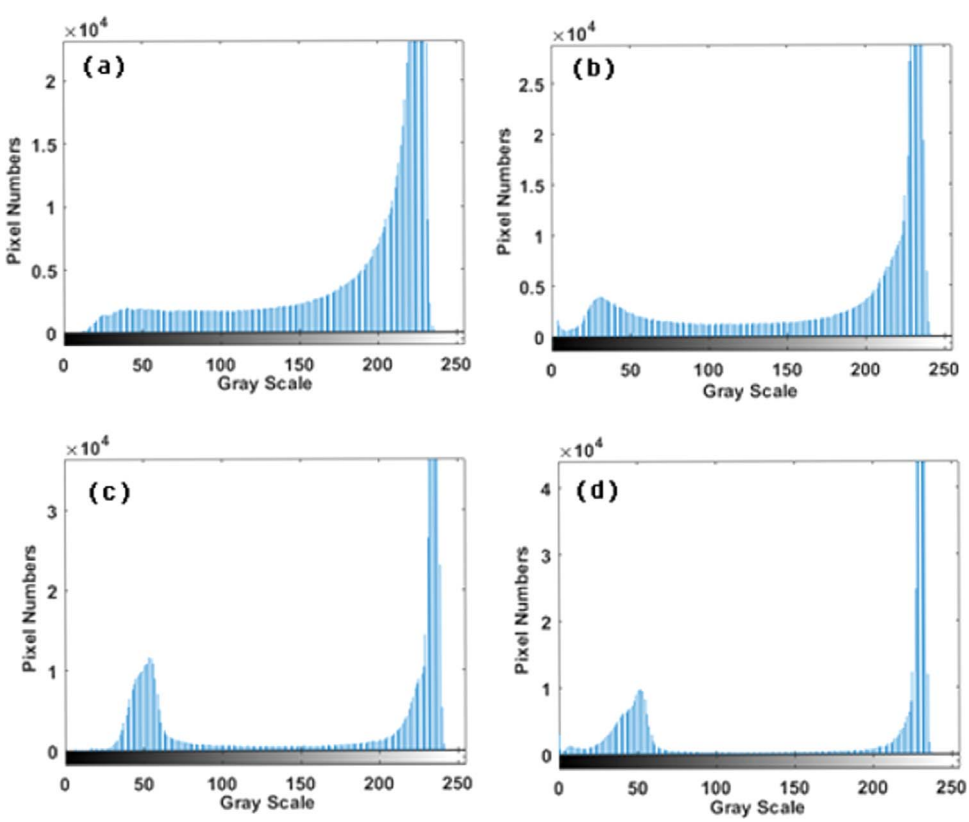

Fig. 3. Grayscale histogram for the microstructures in Fig. 2: (a) WC-1, (b) WC-3, (c) WC-6, and (d) WC-9.

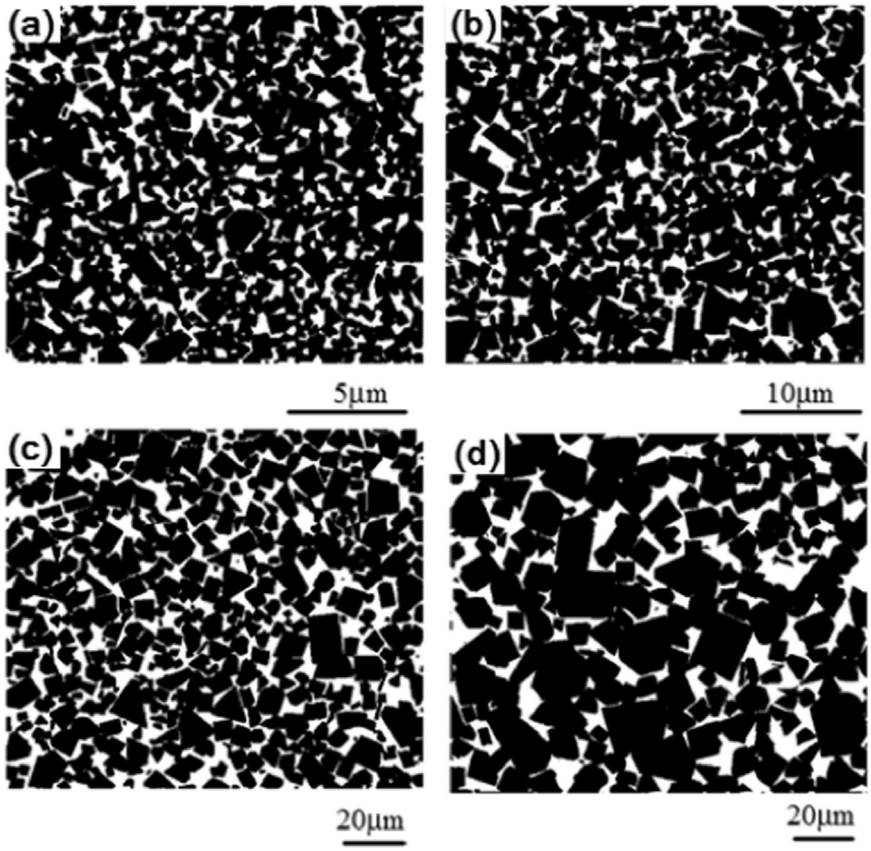

Fig. 4. Binarization of the microstructures in Fig. 2 processed with Otsu's maximum class square error method: (a) WC-1, (b) WC-3, (c) WC-6, and (d) WC-9.

optimum grayscale thresholds of the four images in Fig. 2 were calculated to be 140, 142, 148, and 144, respectively. Fig. 4 shows the images in Fig. 2 after binarization with the optimum grayscale threshold, where WC particles were extracted and converted into black, and the remaining white area is Co phase.

In order to accurately extract the WC/WC grain boundaries, the classical watershed algorithm combined with distance transformation and binary segmentation were made to the binarized microstructure in Fig. 4. Fig. 5 shows the finally processed microstructure with well determined WC/WC grain boundaries, which is ready for the comprehensive microstructure characterization.

3.2. Microstructure characterization

3.2.1. WC grain size

For each WC grain in Fig. 5, both grain area and grain perimeter can be easily measured through image analysis program in this study and equivalent circle diameter (ECD) was used to characterize the WC grain size per the following equation [20,21]:

$$D = \frac{4A}{P} \tag{2}$$

where A is the WC grain area and P is the WC grain perimeter.

Fig. 6 shows both the relative and cumulative frequency distribution of WC grain size for the four carbides in this study. The cumulative frequency curves are the fitted plots based on Rosin-Rammler Distribution Model [26].

In this study, both the number weighted and the area weighted frequency curves for cemented carbides were plotted in Fig. 5. For the number weighted grain size distribution, the relative frequency can be calculated per the principle in [27]:

$$f_{NW}(D + \Delta D) = \frac{\sum_{D}^{D+\Delta D} n_i}{\sum_{0}^{+\infty} n_i} \tag{3}$$

where $f_{NW}(D + \Delta D)$ is the number weighted relative frequency for the grain size range between D and D + ΔD, n_i is the number of grains whose size is within the range of D and D + ΔD, and ΔD is the bin size.

And for the area weighted grain size distribution, the relative frequency can be expressed as [27]:

$$f_{AW}(D + \Delta D) = \frac{\sum_{D}^{D+\Delta D} n_{iA_i}}{\sum_{0}^{+\infty} n_i A_i} \tag{4}$$

where $f_{AW}(D + \Delta D)$ is the area weighted relative frequency for the grain size range between D and D + ΔD, and A_i is the area of the grains whose size is within the range of D and D + ΔD.

For grain size analysis of traditional cemented carbide, the number weighted frequency is always taken as the default assumption so that the area weighted frequency is rarely mentioned or considered. As shown in Fig. 6, the WC grain size distribution of the area weighted frequency plots is significantly different from that of the number weighted frequency plots due to the fact that finer WC grains may contribute less to the total area of WC phase although they may contribute more to the total number.

Fig. 6 shows clearly that both WC-1 and WC-3 have much finer WC grains compared with WC-6 and WC-9. There is significantly higher percentage of coarse grains for the area weighted frequency plot as compared with the corresponding number weighted frequency plot for all the four carbides and the mean WC grain size (d_{50}) derived from the area weighted frequency plot is larger than the one derived from the number weighted frequency plot by at least 42%.

Although it is difficult to see whether number weighted or area weighted grain size distribution has significant advantage, it is necessary to do further investigation on both distributions, which will

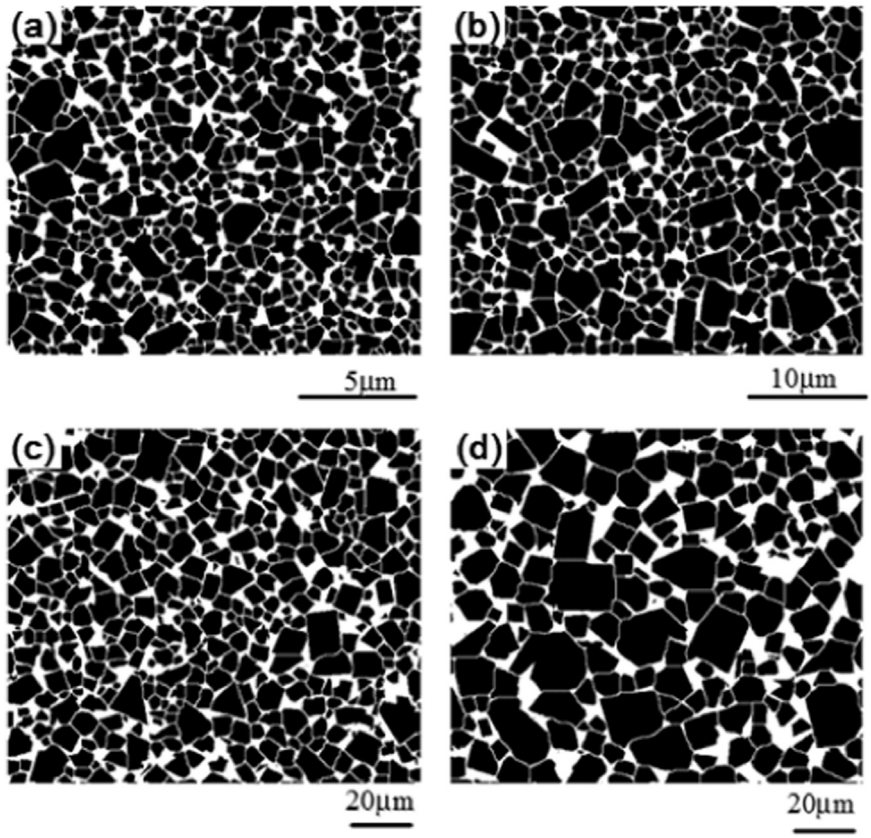

Fig. 5. Microstructure of Fig. 4 with well determined WC/WC grain boundaries through classical watershed process: (a) WC-1, (b) WC-3, (c) WC-6, and (d) WC-9.

definitely affect the entire microstructure evaluation for cemented carbide.

3.2.2. WC/WC contiguity and mean free path of binder

The WC/WC contiguity can be calculated per the following equation for 2D microstructure [13,24]:

$$Cwc/wc = \frac{2\sum_i L^i_{wc/wc}}{\sum_{i,j} 2L^i_{wc/wc} + L^j_{wc/co}} \tag{5}$$

where $L_{WC/WC}$ is the length of WC/WC boundaries, $L_{WC/Co}$ is the length of WC/Co boundaries.

Fig. 7 shows the extracted WC/Co grain boundaries via the binarization in Figs. 4 and 8 shows the extracted WC/WC grain boundaries via the watershed process in Fig. 5. The extracted grain boundaries in both Figs. 7 and 8 can be directly used for WC/WC contiguity calculation in Eq. (5).

Furthermore, the mean free path of Co binder (λ) can be calculated as [13]:

$$\lambda = \frac{d_{WC}(1-f_{WC})}{f_{WC}(1-C_{WC/WC})} \tag{6}$$

where λ is the mean free path of binder, d_{WC} is the mean WC grain size, $C_{WC/WC}$ is the contiguity of WC/WC interface calculated through Eq. (5), f_{WC} is the volume fraction of WC phase in the WC-Co cemented carbide. The mean WC grain size, d_{WC}, can be calculated as d_{50} from the cumulative frequency plots in Fig. 6. Since both the number

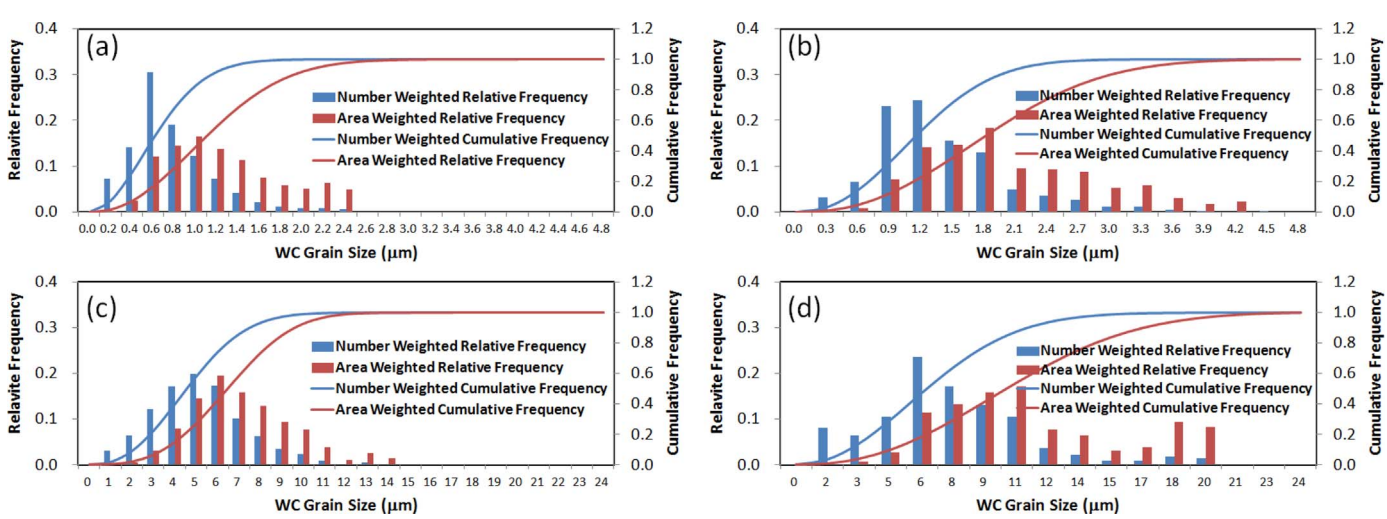

Fig. 6. WC grain size distribution for carbides in this study: (a) WC-1, (b) WC-3, (c) WC-6, and (d) WC-9.

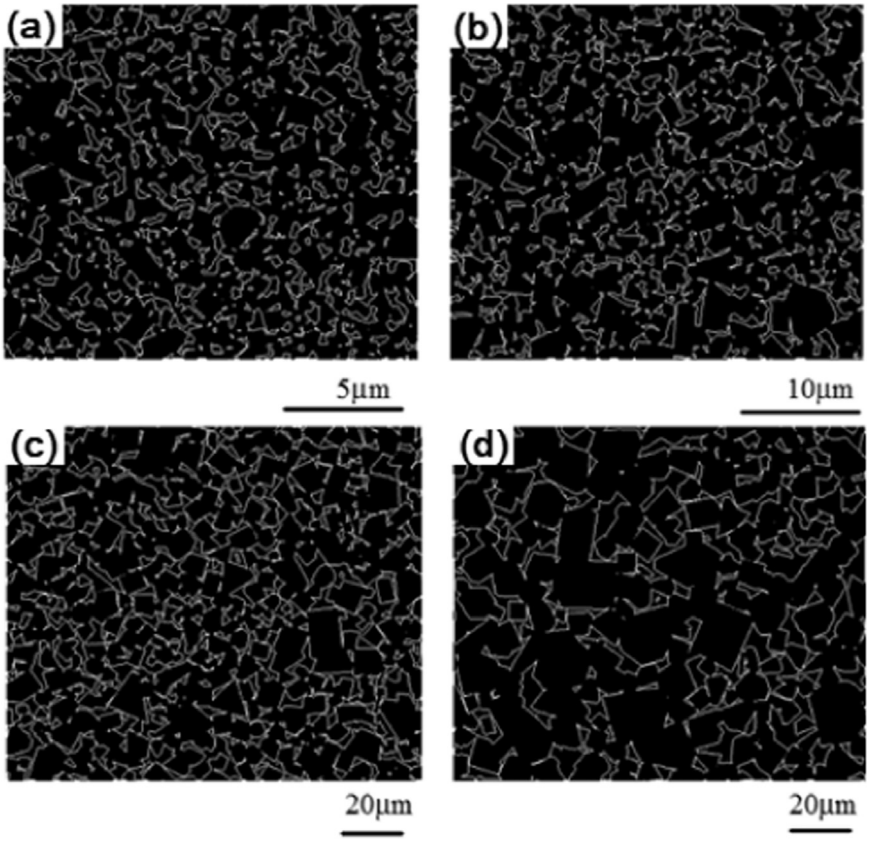

Fig. 7. WC/Co interface extraction based on binary segmentation in Fig. 4: (a) WC-1, (b) WC-3, (c) WC-6 and (d) WC-9.

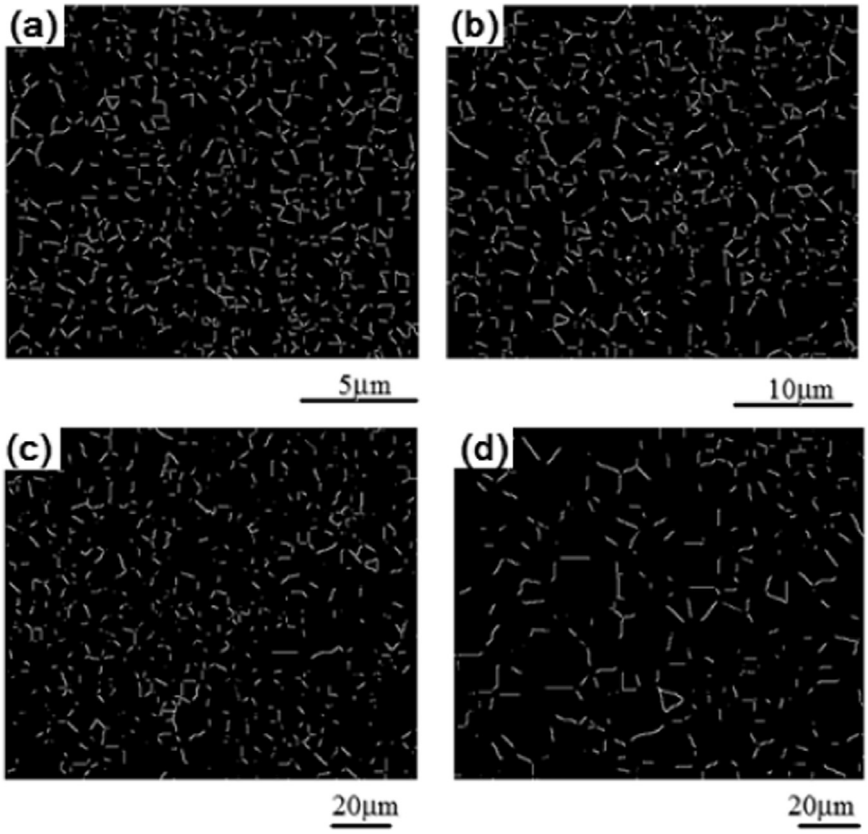

Fig. 8. WC/WC grain boundary extraction from Fig. 5: (a) WC-1, (b) WC-3, (c) WC-6, and (d) WC-9.

Table 2
The measured microstructure parameters and Vickers hardness of the samples in this study.

Sample No.	WC (vol%)	d_{WC} (d_{50-n})	d_{WC} (d_{50-A})	Cwc/wc	λ_n	λ_A	Hardness (HV_{10})
WC-1	78.54	0.61	1.10	0.48	0.32	0.58	1253 ± 10
WC-3	78.42	1.14	1.81	0.48	0.61	0.96	1170 ± 15
WC-6	74.12	4.55	6.46	0.38	2.55	3.62	886 ± 12
WC-9	77.49	6.22	10.03	0.40	3.02	4.87	837 ± 16

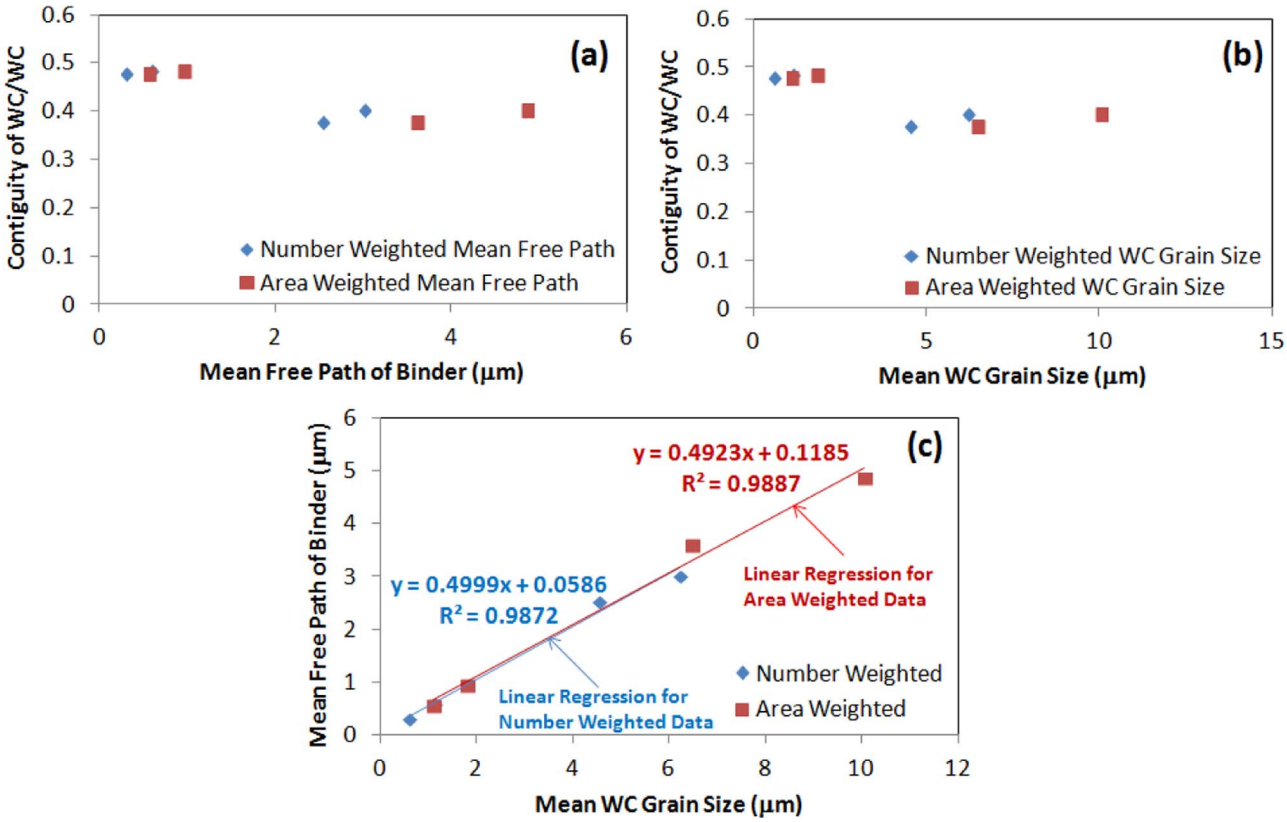

Fig. 9. The relationships of (a) contiguity of WC grains ($C_{WC/WC}$) vs. mean free path of Co binder (λ), (b) contiguity of WC grains ($C_{WC/WC}$) vs. mean WC grain size (d_{WC}), and (c) mean free path of Co binder (λ) vs. mean WC grain size (d_{WC}).

weighted mean WC grain size, d_{50-n}, and the area weighted mean WC grain size, d_{50-A}, can be used as d_{WC} in Eq. (5), the corresponding mean free path of binder will be λ_n and λ_A, respectively.

The measured microstructure parameters and the Vickers hardness for samples WC-1, 3, 6, and 9 are listed in Table 2. And the relationships among the mean WC grain size (d_{WC}), the contiguity of WC grains ($C_{WC/WC}$), and the mean free path of Co binder (λ) are shown in Fig. 9. For both the number weighted and area weighted data, the contiguity of WC grains decreases slightly with either the mean WC grain size or the mean free path of binder and reaches a plateau quickly (Fig. 9(a) and (b)), indicating that there is no strong correlation between the contiguity of WC grains and either the mean WC grain size or the mean free path of binder. Fig. 9(c) shows clearly the linear relationship between mean WC grain size and mean free path of Co for both number weighted and area weighted data, which agree with Eq. (6) very well.

According to Chermont and Osterstock's model [28], both the yield strength as well as the hardness of cemented carbide have a linear relationship with the inverse square root of mean WC grain size. As shown in Fig. 10, for both number weighted and area weighted mean WC grain size, the Vickers hardness is linearly related to the inverse square root of mean WC grain size, which is consistent with Chermont and Osterstock's model. Compared with number weighted mean WC grain size, the Vickers hardness has a better linear relationship with the inverse square root of area weighted mean WC grain size with higher R

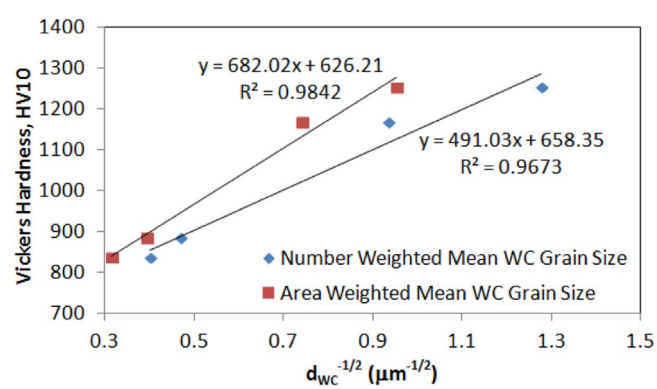

Fig. 10. The Vickers hardness as the function of inverse square root of mean WC grain size for both number weighted and area weighted data.

square. Although it is too early to tell whether number weighted or area weighted mean WC grain size (as well as the mean free path of binder) is better, this study shows that area weighted WC grain size (and the subsequent mean free path of Co phase) is more suitable for Chermont and Osterstock's hardness modeling. The area weighted WC grain size as well as the corresponding mean free path of binder deserve more thorough investigation in order to have a better understanding of

microstructure evolution as well as a better mechanical behavior modeling for cemented carbide.

4. Conclusions

In this study, an automatic image analysis procedure based on maximum class square error and watershed transformation techniques has been successfully developed. This procedure was used for quick and consistent microstructure characterization of WC-16Co with different WC grain sizes. The conclusions include:

1. For both WC grain size distribution as well as the mean WC grain size, the number weighted results are quite different from the area weighted ones. The area weighted data tends to have higher percentage of coarser WC grains and larger mean WC grain size compared with the number weighted data.
2. For both number weighted and area weighted data, the mean WC grain size (d_{WC}) has quite similar linear relationship with mean free path of Co binder (λ).
3. The contiguity of WC grains ($C_{WC/WC}$) varies between 0.38 and 0.48 in WC-16Co carbides, which is not significantly affected by either the mean free path of Co binder or the mean WC grain size.
4. The hardness of WC-Co cemented carbide has linear relationship with the inverse square root of mean WC grain size at the same volume fraction of WC. The hardness shows better linear regression fit with the inverse square root of area weighted mean WC grain size compared with that of number weighted mean WC grain size.
5. Although it is too early to conclude whether the number weighted or the area weighted WC grain size (and corresponding mean free path of Co binder) is better, the testing results in this study shows that the area weighted WC grain size (and corresponding mean free path of Co binder) is more suitable for Chermont and Osterstock's hardness modeling compared with number weighted WC grain size.
6. The area weighted WC grain size and subsequent mean free path of Co binder, which have rarely been considered for microstructure characterization of cemented carbide previously, deserve more thorough investigation and could be the key parameters for a better understanding of the microstructure evolution, as well as a better mechanical behavior modeling for cemented carbide.

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References

[1] P.K. Katiyar, P.K. Singh, R. Singh, et al., Modes of failure of cemented tungsten carbide tool bits (WC/Co): a study of wear parts, Int. J. Refract. Met. Hard Mater. 54 (2016) 27–38.
[2] B. Seo, S. Kim, Cobalt extraction from tungsten carbide-cobalt (WC-Co) hard metal scraps using malic acid, Int. J. Miner. Process. 151 (2016) 1–7.
[3] H. Zhang, G. Zhong, D. Zhao, Study on characterization parameters of ultra coarse grain cemented carbide cutter, Rock Drill. Mach. Pneum. Tools (2016).
[4] Y. Sun, W. Su, H. Yang, et al., Effects of WC particle size on sintering behavior and mechanical properties of coarse grained WC–8Co cemented carbides fabricated by unmilled composite powders, Ceram. Int. 41 (10) (2015) 14482–14491.
[5] B.M. Moshtaghioun, D.G. García, A.D. Rodríguez, et al., High-temperature creep deformation of coarse-grained boron carbide ceramics, J. Eur. Ceram. Soc. 35 (5) (2015) 1423–1429.
[6] H.E. Exner, J. Gurland, A review of parameters influencing some mechanical properties of tungsten carbide–cobalt alloys, Powder Metall. 13 (25) (1970) 13–31.
[7] L.S. Sigl, H.F. Fischmeister, On the fracture toughness of cemented carbides, Acta Metall. 36 (4) (1988) 887–897.
[8] J. Larsen-Basse, Effect of composition, microstructure, and service conditions on the wear of cemented carbides, JOM 35 (11) (1983) 35–42.
[9] K.S. Ravichandran, Fracture toughness of two phase WC-Co cermets, Acta Metall. Mater. 42 (1) (1994) 143–150.
[10] J.W. Lee, D. Jaffrey, J.D. Browne, Influence of process variables on sintering of WC—25 wt% Co, Powder Metall. 23 (2) (1980) 57–64.
[11] H. Engqvist, B. Uhrenius, Determination of the average grain size of cemented carbides, Int. J. Refract. Met. Hard Mater. 21 (1) (2003) 31–35.
[12] S. Lay, J.M. Missiaen, 1.03–Microstructure and morphology of hardmetals, Compr. Hard Mater. 1 (1) (2014) 91–120.
[13] B. Roebuck, E.G. Bennett, Phase size distribution in WC/Co hardmetal, Metallography 19 (1) (1986) 27–47.
[14] I. Borgh, P. Hedström, T. Persson, et al., Microstructure, grain size distribution and grain shape in WC–Co alloys sintered at different carbon activities, Int. J. Refract. Met. Hard Mater. 43 (2014) 205–211.
[15] K. Brookes, Grain-size distribution in WC and WC/Co, Met. Powder Rep. 67 (2) (2012) 30–32.
[16] K. Bonny, P. De Baets, W. Ost, et al., Influence of electrical discharge machining on the reciprocating sliding wear response of WC-Co cemented carbides, Wear 266 (1) (2009) 84–95.
[17] M.A. Yousfi, Evolution of WC-Co Cemented Carbide Microstructure During Creep Testing, 2014.
[18] J.M. Tarrago, D. Coureaux, Y. Torres, et al., Implementation of an effective time-saving two-stage methodology for microstructural characterization of cemented carbides, Int. J. Refract. Met. Hard Mater. 55 (2016) 80–86.
[19] X. Zhang, Analysis of hard-metal grain size base on ImageJ, Mech. Manag. Dev. (2015).
[20] Y.G. Yuan, X.X. Zhang, J.J. Ding, et al., Measurement of WC grain size in ultrafine grained WC-Co cemented carbides, Appl. Mech. Mater. 278 (2013) 460–463 Trans Tech Publications.
[21] S.J. Park, K. Cowan, J.L. Johnson, et al., Grain size measurement methods and models for nanograined WC–Co, Int. J. Refract. Met. Hard Mater. 26 (3) (2008) 152–163.
[22] X.U. Jian-Lin, L.U. Yang, L.I. Wen-Sheng, et al., Application of stereology and image analysis technology in materials science, Jinshu Rechuli/Heat Treat. Met. 30 (4) (2005) 7–10.
[23] G. Liu, Applied stereology in materials science and engineering, J. Microsc. 171 (1) (1993) 57–68.
[24] G.S. Upadhyaya, Cemented Tungsten Carbides: Production, Properties and Testing, Noyes Publications, Westwood, New Jersey, USA, 1998.
[25] N. Otsu, A threshold selection method from gray-level histograms, IEEE Trans. Syst. Man Cybern. 9 (1) (1979) 62–66.
[26] A. Macıas-Garcıa, E.M. Cuerda-Correa, M.A. Dıaz-Dıez, Application of the Rosin–Rammler and Gates–Gaudin–Schuhmann models to the particle size distribution analysis of agglomerated cork, Mater. Charact. 52 (2) (2004) 159–164.
[27] L. Wojnar, K.J. Kurzydlowski, J. Szala, Quantitative image analysis, in: G.F. Vander Voort (Ed.), Metallography and Microstructures, ASM Handbook Vol. 9, ASM International, Materials Park, OH, 2004, pp. 403–427.
[28] J.L. Chermant, F. Osterstock, Elastic and plastic characteristics of WC-CO composite materials, Powder Metall. Int. 11 (3) (1979) 106–109.